



CATALYST SOLUTIONS PTE LTD

Agentic Bid & Tender Intelligence

Catalyst_Tender_AI — an autonomous, gated tender lifecycle platform built for large EPC and infrastructure contractors.

Nine-stage lifecycle

Ten specialist agents

Three human gates

One source of truth

What we will cover



The tendering challenge

Where large EPC bids lose time, margin and win rate today

01



Introducing Catalyst_Tender_AI

An orchestrated mesh of ten agents over a single source of truth

02



How the platform works

Nine stages, three human gates, provider-agnostic architecture

03



What changes in your business

Efficiency, accuracy, vendors, procurement, margin, delivery

04



Delivery and next steps

Six months to production go-live, and how we start

05

Tendering is where EPC contractors win or lose the year

Every major bid is a compressed, document-heavy, multi-party programme run against an immovable deadline.

A single ITT can arrive as hundreds of pages across specifications, drawings, forms and addenda. From that, a bid team must extract every obligation, package the scope, price dozens of subcontract trades, build a credible programme, draft a persuasive proposal and evidence total compliance — usually in three to six weeks, alongside every other live bid.

The work is done well by experienced people. It is simply done manually, in parallel, under time pressure, and largely without institutional memory.

The bid team is not the bottleneck. The process around them is.



Fragmented intake

Portals, inboxes and hard copy monitored inconsistently — tenders found late or missed entirely



Manual re-keying

Requirements transcribed by hand into matrices and trackers, with extraction error



Chase-and-nudge

Buyers spend their week pursuing subcontractor quotes rather than evaluating them



Scattered cost models

Spreadsheets with inconsistent treatment of overheads, escalation and bonds

Four pressures are converging on the bid office

Each of them raises the cost of tendering manually — and the cost of getting it wrong.



Rising bid volume and complexity

More opportunities, more addenda, longer requirement sets and tighter submission windows — with the same bid resource.



Margin compression

Competitive pricing meets volatile material, FX and escalation exposure. Pricing on instinct is no longer survivable.



Compliance and ESG burden

Regulatory, QHSE and ESG evidence requirements expand every cycle, and a single mandatory gap disqualifies the bid.



Scarce estimating talent

Senior estimators and proposal managers are hard to replace, and their knowledge leaves with them when they go.

01

Where tendering breaks down

Nine stages, nine distinct failure modes — and they compound.

Intake and screening · Subcontractor orchestration · Bid/No-Bid judgement · Pricing and margin · Compliance and submission

Nine points of failure across the tender lifecycle

Mapped to the nine stages every EPC bid actually passes through.

1 Identification & screening

Portals watched inconsistently; tenders surface late, leaving no time to shape the bid.

2 Subcontractor inputs

Quotes arrive in incomparable formats, late, or not at all; coverage is thin on key packages.

3 Bid / No-Bid

Pursuit decided on anecdote and optimism; weak bids consume the resource strong bids needed.

4 Scheduling & planning

Programme built in isolation from cost; it stops reflecting reality the moment scope moves.

5 Cost & margin modelling

Overheads, bonds, escalation and FX treated inconsistently between bids and estimators.

6 Proposal drafting

Content re-invented from scratch; what won last time is not found, so it is not reused.

7 Compliance verification

The matrix is assembled last, under pressure, so gaps are found far too late to close well.

8 Compilation & submission

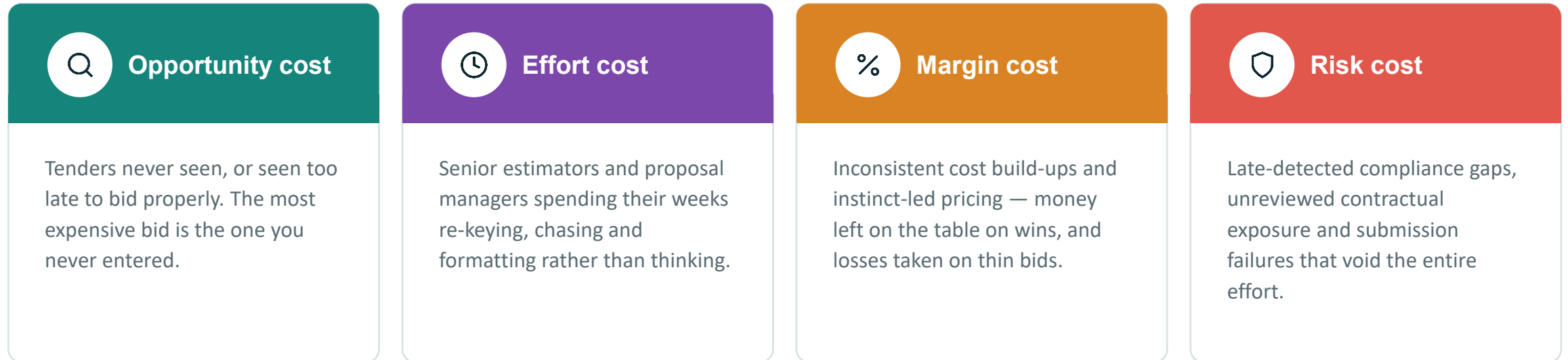
Formatting, signatures and portal constraints become a deadline-night risk to the whole bid.

9 Post-award & learning

Outcomes are recorded but never analysed; the next bid starts no wiser than the last.

What manual tendering actually costs you

Four cost buckets, only one of which usually appears in a budget line.



Only effort cost is visible on a budget. The other three are absorbed silently into win rate and delivered margin.

Vendor management is a chase, not an evaluation

Procurement capacity is consumed by pursuit — leaving little left for the judgement that adds value.

A buyer's week today

- Rebuild the scope split for every new tender from scratch
- Assemble a supplier list from memory, email history and habit
- Issue RFQs manually, package by package, with hand-picked attachments
- Chase non-responders by phone and email, repeatedly
- Re-type incoming quotes into a comparison sheet to make them comparable
- Discover exclusions and validity gaps only when the numbers refuse to reconcile

The consequences

-  **Thin coverage** Two quotes where five were needed — so the best price was never seen.
-  **Weak comparability** Offers with different exclusions and validity compared as if identical.
-  **Supplier fatigue** Good subcontractors deprioritise a buyer whose RFQs are late and incomplete.
-  **No memory** Delivered performance never feeds back into who gets asked next time.

Margin is decided in spreadsheets nobody can reconstruct

The commercial heart of the bid is the least governed part of the process.



Inconsistent build-ups

Every estimator layers overheads, preliminaries, insurances and bonds slightly differently. Two bids for similar work carry different embedded margin, and nobody can see it.



Static assumptions

FX, escalation and material exposure are fixed at a point in time. When a supplier re-quotes in week four, the model is not re-run — it is patched.



No scenario view

One number goes to the review board. There is no base, stretch and defensive case, no sensitivity, and no explicit break-even.



Instinct-led pricing

The final margin call is made on experience and pressure rather than on evidence of what actually wins at what price.

The compounding effect

Because the cost model cannot be reconstructed, it cannot be challenged — and because it is never compared against as-built outcomes, it never improves.

The result is a pricing function that carries the largest commercial risk in the business while generating the least evidence about its own accuracy.

Compliance is verified last — when there is no time to fix it

The final week concentrates the highest-consequence work into the smallest available window.



A single missed mandatory requirement can void a bid that consumed months of effort and hundreds of thousands in bid cost. The failure is almost never capability — it is sequence.

What a governed tender operation looks like

The target state is not a faster manual process — it is a different operating model.

TODAY

- ⊗ Tenders found when someone remembers to look
- ⊗ Requirements re-keyed by hand
- ⊗ Pursuit decided on instinct
- ⊗ Buyers chasing quotes
- ⊗ Cost models nobody can reconstruct
- ⊗ Compliance checked at the end
- ⊗ Outcomes filed and forgotten

WITH CATALYST_TENDER_AI

- ✓ Every configured portal and mailbox monitored continuously, 24/7
- ✓ Structured extraction against a fixed schema, validated by a coordinator
- ✓ Win-probability scored on history, with an explicit uncertainty band
- ✓ RFQs issued, tracked and nudged automatically; quotes normalised on arrival
- ✓ Traceable build-up with base, stretch and defensive margin scenarios
- ✓ Requirement-to-evidence matrix maintained continuously, gated before submission
- ✓ Every win and loss refines scoring, supplier ranking and margin models

02

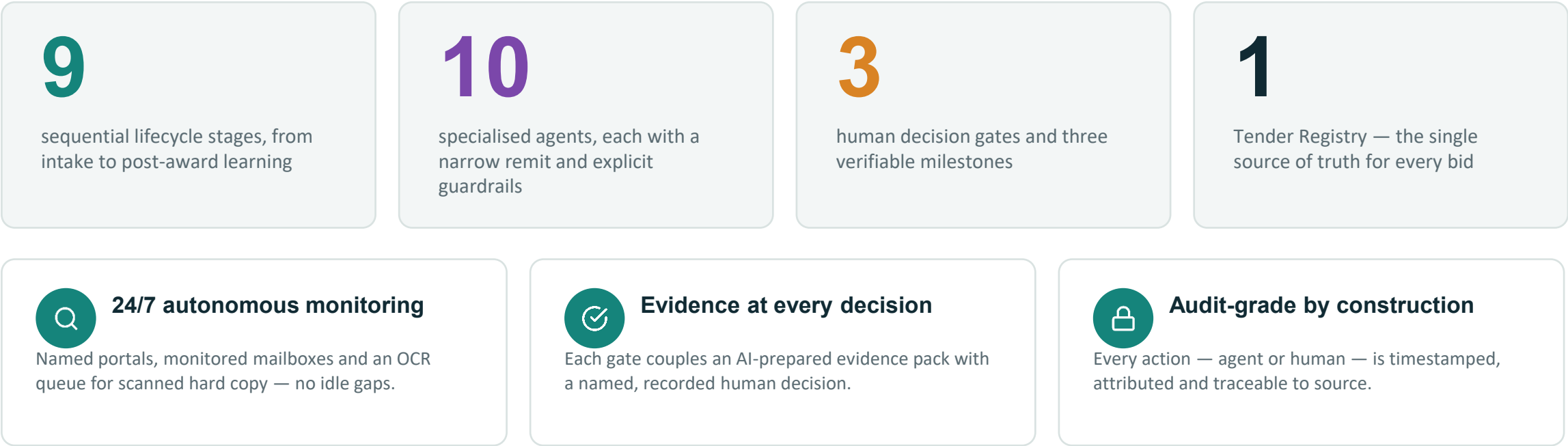
Introducing Catalyst_Tender_AI

An orchestrated mesh of ten specialised agents working over a single source of truth.

Automation by default · Human judgement at the gates · Total auditability · Continuous learning

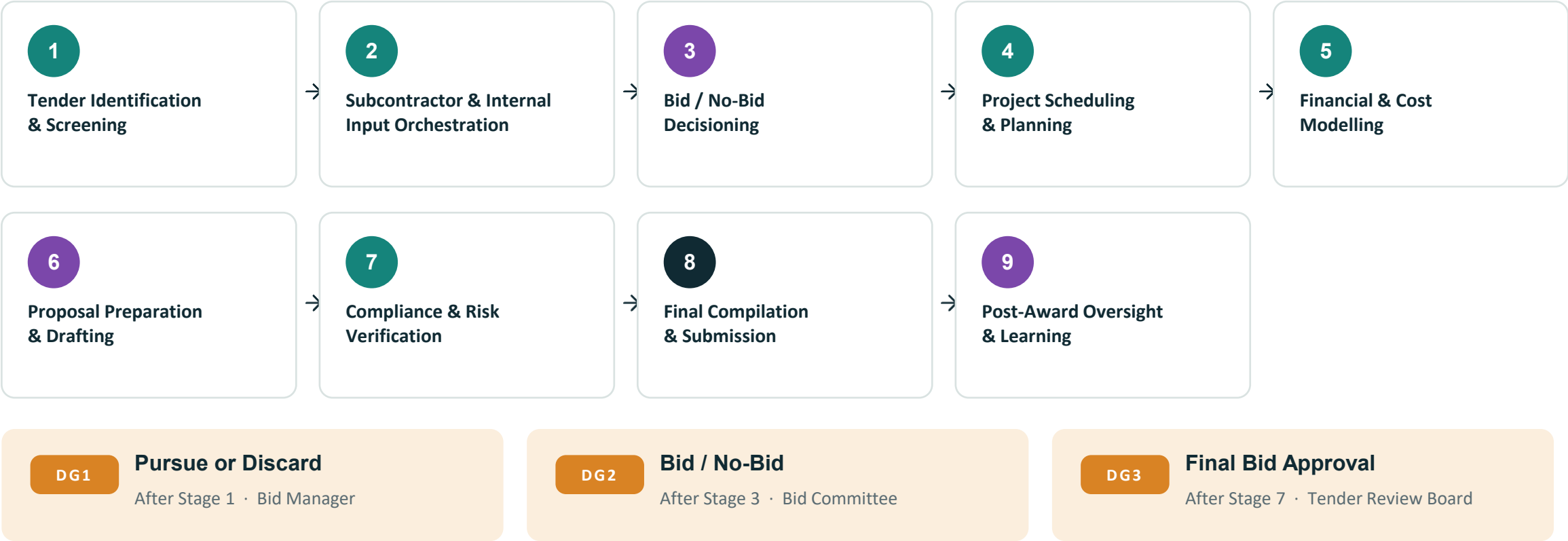
Catalyst_Tender_AI at a glance

Catalyst_Tender_AI replaces manual tender friction with an orchestrated mesh of specialised AI agents working over one authoritative registry. Agents monitor, extract, orchestrate, draft, verify and submit. People decide — at three mandatory gates that cannot be bypassed.



The nine-stage agentic workflow

Stages 4 and 5 run in parallel; Stage 9 feeds every earlier stage through the continuous learning loop.



Ten specialised agents, each with a narrow remit

Every agent has defined inputs and outputs, explicit guardrails, evaluation thresholds and a named human handoff.

 Intake & Extraction Tender Registry record with structured requirement set	S1	 Outreach & Evaluation Normalised quote register and best-fit combinations	S2
 Win-Probability & Recommendation Win-probability score with confidence band	S3	 Scheduling Baseline Gantt, critical path and resource histogram	S4
 Costing & Margin Priced BOQ with cost build-up and margin scenarios	S5	 Drafting & Section Assembly Technical, commercial and compliance drafts	S6
 Compliance Verification Compliance matrix at 100% mandatory coverage	S7	 Document Assembly & e-Submission Submitted bid package with acknowledgement	S8
 Delivery Oversight Live KPI dashboards and corrective actions	S9A	 Learning Loop Refreshed win, supplier and margin models	S9B

Automation by default, humans at the decision gates

Agents are accountable for throughput and quality of work product. People remain accountable for judgement.

What the agents never do

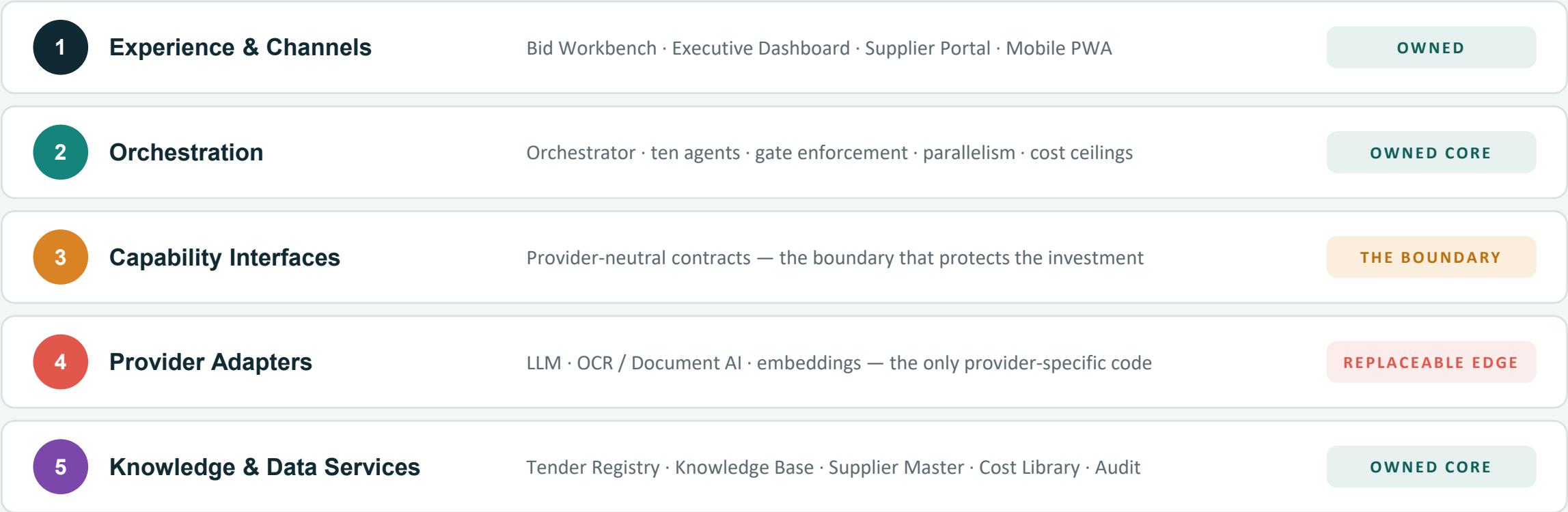
-  The Intake Agent never discards a tender — it only flags for a human decision
-  The Win-Probability Agent never makes the Bid / No-Bid call
-  The Costing Agent never submits a price without Commercial Manager selection
-  The Outreach Agent recommends only — it never issues a commitment or PO
-  The Document Assembly Agent never submits without an authorised signatory

DG1	Pursue or Discard	Bid Manager	≤ 24h
DG2	Bid / No-Bid	Bid Committee / Sponsor	≤ 24h
DG3	Final Bid Approval	Tender Review Board	≤ 48h
M1	Tender Logged & Qualified	Tender Coordinator	Stage 1
M2	Baselines Approved	Planning + Commercial	Stages 4–5
M3	Bid Submitted	Bid Manager	Stage 8

Gates cannot be bypassed — the orchestrator blocks progression until each decision is recorded with its rationale.

A five-layer platform with a hard capability boundary

The bulk of the engineering effort and all of your data sit in a stable, owned core.



Sources feed in at the top — portals, mailboxes, scanned documents, supplier submissions and the past-bid library. Governed outputs flow out: submissions, dashboards and telemetry.

Provider-agnostic by design — you are never locked in

The reasoning model is rented inference reached across a stable internal boundary. Everything of value is yours.

STABLE CORE — OWNED

- Agent orchestration and workflow logic
- Prompt templates as versioned, model-tagged assets
- The retrieval pipeline, indices and knowledge base
- Tool definitions and every integration they call
- Business rules, guardrails and the audit trail
- The evaluation harness and all accumulated data

REPLACEABLE EDGE — RENTED

- The reasoning model, reached only through the Model Gateway
- The OCR and document-AI engine, reached through an adapter
- The embedding and reranking provider
- Selected by configuration, never by a code path
- Routed per task and cost across frontier, workhorse and economy tiers
- Switched with automatic failover on outage or rate limit



Changing model provider is a configuration and revalidation exercise of two to four weeks — never a rebuild. That ability is your strongest long-term cost-control mechanism, and it is designed in from day one.

Security, data sovereignty and audit-grade governance

Tender data is among the most commercially sensitive information your business holds. It is treated that way.



Identity & access

Single sign-on with MFA, role-based access mapped to the RACI, and a scoped service identity for every agent.



Data protection

TLS 1.2+ in transit and AES-256 at rest, field-level masking, and key management through your corporate KMS.



Data residency

Residency aligned to regulation and configured per deployment; cross-border flows reviewed and approved.



AI-specific controls

Prompt-injection defences, tool-use allow-listing, egress allow-listing and PII redaction before any external model call.



Auditability

Every action timestamped, attributed and immutable, with single-click lineage from a bid sentence to its source.



AI governance

A governance forum approves model changes; releases are gated on evaluation pass-rates with canary and rollback.

It works with the systems you already run

Catalyst_Tender_AI is an orchestration layer over your landscape, not a replacement for it.



ERP / Finance

Supplier master, costs and financial actuals



CRM

Client and opportunity context



Procurement / e-Sourcing

RFQ dispatch and supplier qualification



Planning tool

Baseline schedules and resource library



Document repository

All bid artefacts, under version control



Email / calendar

Tender intake and deadline tracking



Identity provider

Single sign-on and role-based access



e-Submission portals

Bid submission and acknowledgement capture



Analytics / warehouse

Learning loop and outbound metrics

Five integration patterns are used as appropriate: synchronous APIs, event-driven flows, scheduled batches, webhooks, and document-based exchange with provenance tracking where no API exists.

03

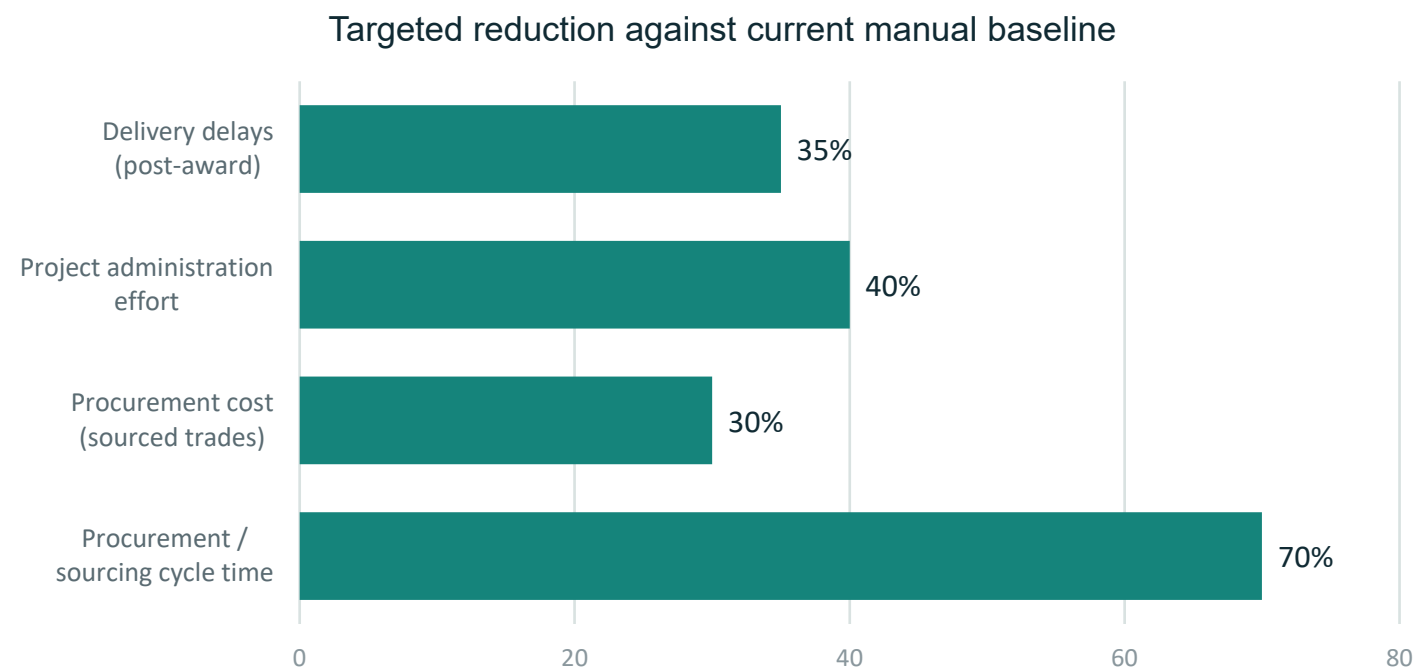
What changes in your business

Six operating domains, each with a measurable target and a named mechanism behind it.

Efficiency · Accuracy · Vendor management · Procurement · Margin control · Delivery assurance

The bid machine runs at machine speed

Engineered targets from the system specification, measured and recalibrated quarterly.



≤ 15 min

from tender receipt to logged in the registry, against days of manual screening

≤ 24 hrs

for 100% of RFQs to be issued to shortlisted subcontractors after DG1

24 / 7

autonomous monitoring of every configured portal, mailbox and OCR queue

Initial screening compresses from days to minutes, so bid teams spend the recovered time on strategy rather than administration.

Right first time, and provably so

Accuracy is engineered through structured extraction, mandatory verification and an immutable evidence trail.

≥ 95%

extraction accuracy on key fields —
deadline, scope and eligibility

100%

compliance matrix coverage of
mandatory tender requirements

Zero

missed tenders from configured
sources, verified by daily
reconciliation

100%

of actions traceable — full audit-trail
coverage across all stages

How the numbers are actually achieved



Schema-bound extraction

Every requirement is extracted into a machine-readable record against a fixed schema — never free-typed into a tracker.



Deduplication and versioning

One tender resolves to exactly one registry record, with addenda linked to their parent so nothing is priced twice or missed.



Confidence routing

Any field below the confidence threshold is routed to the Tender Coordinator for review; it is never silently accepted.



Independent verification

A reviewer outside the drafting team runs a mandatory integrity pass before the Tender Review Board sees the pack.

From chasing subcontractors to selecting them

The Outreach & Evaluation Agent runs the entire supplier engagement cycle, and remembers how each one performed.

30–50%

targeted improvement in bid coverage and supplier quality

Guardrails

- Anti-bribery and sanctions screening before engagement
- Fair, criteria-based shortlisting with no unexplained preference
- Recommends only — never issues a commitment or PO
- Confidentiality of competing quotes preserved

Criteria-based shortlisting

Prequalified subcontractors ranked per package on performance history, capability fit and current capacity.

Automated RFQ issuance

RFQs with the correct scope extracts, drawings and forms issued to every shortlisted supplier via email or portal.

Tracked responses and nudges

Acknowledgements and submissions tracked; non-responders nudged automatically within a configured SLA.

Quote normalisation

Offers parsed into a common schema — unit price, lump sum, validity, exclusions and lead times — so they are genuinely comparable.

Best-fit evaluation

Combinations recommended on price, capability and risk, with the buyer free to override any recommendation.

Performance feedback loop

Delivered performance updates supplier scoring, so the next shortlist is better than the last.

Sourcing becomes a governed, evidence-based process

Faster sourcing cycles, better coverage and a defensible record behind every award recommendation.



Up to 70%

reduction in procurement and sourcing cycle time



Up to 30%

reduction in procurement cost on sourced trades



One

supplier master shared across ERP and procurement

What procurement gains



Scope packaged consistently

Civil, MEP, specialist trades, equipment and services broken out the same way on every bid.



Wider, faster market coverage

More suppliers approached per package in less time, so the competitive field is genuinely tested.



Clarifications closed, not carried

Missing data requested automatically and tracked until closed — zero stale items at stage exit.



Evidence behind every award

The full shortlist, quotes, normalisation and rationale retained against the tender record.

Pricing becomes defensible, dynamic and comparable

The Costing & Margin Agent builds a traceable cost model and projects the margin scenarios your board should be choosing between.

BASE

The expected case, built from firm subcontractor pricing and historical productivity norms.

STRETCH

The competitive case, with the sensitivity and break-even analysis that shows where it stops being viable.

DEFENSIVE

The protected case, where risk, escalation and contingency exposure are fully priced in.

Structured cost build-up

Priced subcontractor packages, materials and labour aggregated into a BOQ with a clear direct and indirect split.

Complete indirect layer

Site overheads, project management, insurances, bonds, contingency and head-office recovery layered consistently every time.

FX, escalation and tax

Multi-currency exposure, escalation provisions and duty treatment computed rather than assumed.

Live re-pricing

When a supplier re-quotes or market data moves, the model re-runs — it is not patched by hand.

Cashflow visibility

A bid-stage cashflow profile flags working-capital implications before the price is committed.

Calibrated against outcomes

Margin variance is tracked against as-built history, so the model measurably improves bid on bid.

The Commercial Manager still selects the strategy. The platform makes sure the choice is informed, traceable and reproducible.

Submit on time, then deliver what you promised

The bid does not end at submission — commitments made in the proposal become tracked obligations in delivery.

STAGE 8 — SUBMISSION



100% on-time submission

Deadline-aware assembly with a secondary-channel fallback if a portal fails.



Multi-format generation

Word, PDF, Excel and JSON produced to the tender’s own templates and format rules.



Signatures and attachments

Certificates, bonds, guarantees and prequalification documents placed automatically.



Receipt captured every time

Submission receipts time-stamped and stored against the tender record.

STAGE 9 — POST-AWARD & LEARNING



~35% fewer delivery delays

Milestones, obligations and resource updates tracked live, with teams nudged before slippage becomes real.



Margin variance in view

Delivered margin measured against the bid commitment, continuously rather than at year end.



Full traceability at handover

Schedule, cost model, subcontractor commitments and risks handed to the Project Director intact.



Every outcome teaches the system

Wins and losses refine win-probability scoring, supplier rankings and margin models quarter on quarter.

SUMMARY

The outcome scorecard

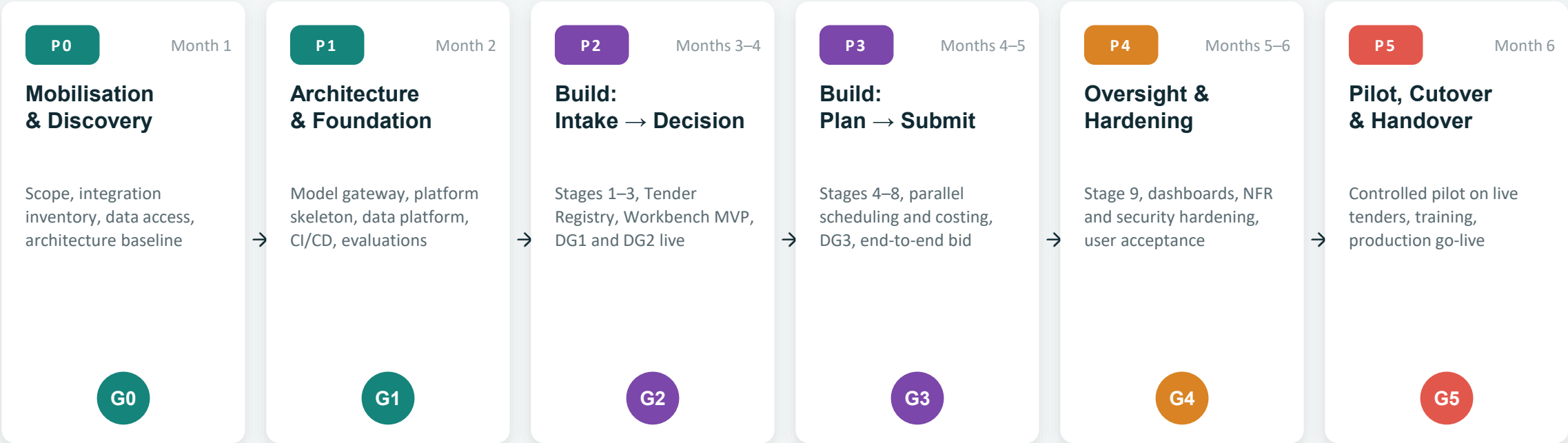
The measurable promise of the platform, and the basis on which benefits realisation is tracked.

● Procurement / sourcing cycle time	Up to 70% reduction	Stages 1, 2, 9	● Tender intake to logged in registry	≤ 15 minutes	Stage 1
● Bid coverage & supplier quality	30–50% improvement	Stage 2	● Procurement cost (sourced trades)	Up to 30% reduction	Stages 2, 5
● Extraction accuracy on key fields	≥ 95%	Stage 1	● Compliance matrix coverage	100% of mandatory items	Stage 7
● On-time submission rate	100%	Stage 8	● Win-probability calibration vs actual	Within ±10%	Stages 3, 9
● Project administration effort	25–40% reduction	Stages 6, 9	● Delivery delays (post-award)	~35% fewer	Stage 9
● Audit-trail coverage	100% of actions traceable	All stages			

Targets are reviewed quarterly and recalibrated by the Bid Office with the Executive Sponsor. Every one of them is instrumented in the platform.

Six months from mobilisation to production go-live

MVP-first: a usable intake-to-decision slice by month four, a full end-to-end bid by month five.



Continuous workstreams:

- Evaluation & regression harness
- Security, AI governance & compliance
- DevOps, CI/CD & observability
- Integration engineering

Built for how EPC contractors actually tender



Domain-built, not adapted

Nine stages, three gates and ten agents modelled on the real EPC bid lifecycle — including the parallel scheduling and costing branch.



Governed and audit-grade

Human accountability preserved at every gate, with an immutable trail from any bid sentence back to its source.



Yours to keep

Orchestration, prompts, retrieval, data and evaluations are owned IP. The model layer is replaceable by configuration.

Next steps

01

Discovery workshop

A half-day session with your bid office to map the nine stages onto your current process.

02

Tender data assessment

We review a sample of past bids to size the knowledge base and bootstrap the evaluation set.

03

Proof of value

A scoped pilot on Stage 1 intake and screening against your live portals and mailboxes.

04

Programme proposal

A costed six-month delivery plan with named team, gates and acceptance criteria.